
Handwritten Digit Recognition Using HOG, Zoning, and Projection-Based Feature Extraction

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Abstract

This paper presents an image processing pipeline for handwritten digit recognition, based on interpretable feature extraction techniques. The method used in this paper relies on handcrafted features such as Histogram of Oriented Gradients (HOG), zoning-based spatial summaries, and projection histograms to capture both local gradient structure and global spatial information. A Random Forest Classifier was used to evaluate the effectiveness of these features. The experiments were based a dataset containing normalized handwritten digits, automatically scanned from envelopes by the U.S. Postal Service. The experiments found that zoning had little effect on the accuracy, however projection histograms yielded significant improvements in accuracy, achieving up to 89.69% on the test set. An analysis of the confusion matrix revealed certain several misclassified digit pairs. To address this, a combination of horizontal and vertical projection histograms were introduced, which improved the accuracy of confused classes considerably.

1 Introduction

Handwritten digit recognition is a machine learning task that involves identifying numeral digits from images of handwritten text. This is a complex pattern recognition problem due to the variability of human handwriting.

In this paper, we explore the task of handwritten digit classification using classical image processing and feature engineering techniques. The goal is to create and evaluate a pipeline that yields a high accuracy without relying on convolutional neural networks. In this paper, we design an interpretable and modular system using feature extraction methods such as Histogram of Oriented Gradients (HOG), zoning, and projection histograms.

The dataset used consists of 16x16 grayscale images of handwritten digits, normalized and scanned from envelopes by the U.S. Postal Service. Our goal is to first create a preliminary pipeline consisting of just HOG features and a Random Forest Classifier and use the evaluation results to determine what may be potentially good features to add. Through our experiment, we aim to answer the following questions:

1. Which digit classes are most frequently confused with others?
2. Can we design a lightweight and interpretable pipeline that achieves a high accuracy?
3. How do feature combinations affect per-class accuracy and confusion?

Our final pipeline integrates HOG, Zoning, and projection histograms features, which increases overall test accuracy from 86.35% to 89.69%.

2 Methodology

Our approach to handwritten digit recognition is based on a pipeline consisting of three main components: preprocessing, feature extraction, and classification. We test 4 pipelines with varying architectures and hyperparameters concurrently and evaluate their contribution to the model's performance.

2.1 Preprocessing

Our training set contains 7291 training observations and 2007 test observations that have been deslanted and size normalized. Each image in the dataset is represented as a 256-dimensional vector corresponding to a 16×16 grayscale image. Each 256-dimensional input vector was reshaped into 2D arrays representing a 16×16 image. This allows us to process the data as 2D images, preserving spatial relationships between pixels.

2.2 Feature Extraction

Our pipelines had varying feature extraction techniques that contributed to model accuracy. Our first pipeline consisted of only using HOG as features. The algorithm we implemented in this experiment for extracting the HOG features is as follows:

2.2.1 HOG Feature Extraction Algorithm (Pseudocode)

1. Divide the 16×16 input image into non-overlapping cells of size 4×4 .
2. For each pixel in the image, compute the gradient magnitude and orientation using a numerical approximation of derivatives via a $\begin{bmatrix} 1 & -1 \end{bmatrix}$ convolution filter.
3. For each 4×4 cell:
 - (a) Quantize orientation into 9 bins, representing gradient directions from 0° to 180° .
 - (b) Accumulate gradient magnitudes into the corresponding bins.
4. Flatten all cell histograms into a single feature vector.

2.3 Zoning

Our three other pipelines also incorporated zoning features in addition to the HOG features. Zoning partitions the image into a coarse grid (e.g., 2×2 or 4×4), and computes the mean pixel intensity in each zone. These values represent the spatial distribution of ink and are useful for understanding which zones the ink is most concentrated.

2.4 Projection Histograms

Projection Histograms were found to improve our accuracy by 3.24%. We compute horizontal and vertical projection histograms by summing pixel values across rows and columns, respectively. This produces two vectors of length 16 each, which capture the density of ink along each axis. This provides global spatial information important for distinguishing between structurally similar digits such as '3', '5', and '8'.

2.5 Random Forest Classifier

We implemented a Random Forest Classifier as the classification model for this experiment. Random Forest is an ensemble method that constructs multiple decision trees during training and averages the predictions on the trees.

The hyperparameters used for our random forest are as follows:

- Number of trees: 10
- Maximum depth: 15
- Minimum samples per split: 2

The custom implementation selects split thresholds using percentiles of unique feature values. Each tree is trained on a bootstrap sample, and prediction is done by majority vote.

2.6 Pipeline Variants

We evaluated the following feature combinations:

1. **HOG only:** Baseline pipeline with HOG features only.
2. **HOG + Zoning:** Added 4×4 zoning features to assess spatial distributions
3. **HOG + Dense Zoning:** Used 2×2 zoning to test low-resolution spatial info.
4. **HOG + Zoning + Projection Histograms:** Our final and best-performing pipeline, combining all feature types.

Each pipeline is trained on 7291 images and evaluated on a set of 2007 test images.

3 Implementation & Experimentation

3.1 Dataset Description

The dataset used in this project consists of grayscale images of handwritten digits, originally collected by the U.S. Postal Service. Each image has been deslanted and normalized to a fixed size of 16×16 pixels. The dataset is provided in a flattened format, where each image is represented as a 256-dimensional vector followed by a class label (0–9). For feature extraction, we reshaped each vector back into its 16×16 2D form to preserve spatial structure.

The dataset is split into:

- **Training set:** 7291 images
- **Test set:** 2007 images

3.2 Performance Measurement

To evaluate model performance, we use:

- **Overall accuracy:** the percentage of correctly predicted labels over all test samples.
- **Per-class accuracy:** the accuracy for each digit class, computed as the number of correct predictions for a class divided by the total number of images in that class.
- **Confusion matrix:** a 10×10 matrix that shows prediction counts per actual vs. predicted class. This is useful for identifying specific digit confusions.
- **Classification report:** includes precision, recall, and F1-score for each class.

These metrics are computed using `scikit-learn`'s `accuracy_score`, `confusion_matrix`, and `classification_report` functions.

3.3 Validation Process

The validation setup uses a **hold-out evaluation** strategy using the provided dataset split. We train our model on the training set (7291 samples) and evaluate them on the test set (2007 samples) which is never seen by the model. This ensures that our reported results reflect a generalized performance on unseen data.

3.4 Results

We evaluated multiple feature configurations to assess their impact on classification performance. Below are the test accuracies and insights from the experiments:

- **HOG only:** Using Histogram of Oriented Gradients as the sole feature yielded an accuracy of 86.65%.

- **HOG + Zoning:** Adding 4×4 zoning features resulted in a +0.74% accuracy improvement.
- **HOG + Dense Zoning:** Using 2×2 zoning resulted in a -0.25% accuracy decrease.
- **HOG + Zoning + Projection Histograms:** This final configuration achieved the best performance, with an accuracy of 89.69%, which is a 3.04% increase from the original. It significantly improved recognition of previously confused digits such as 2, 4, and 8.

3.5 Baseline Pipeline Results

Our baseline pipeline which consisted of only the HOG features and Random Forest Classifier received the following results:

3.5.1 Accuracy

Overall Accuracy: 0.8665

Per-Class Accuracy:

Class 0: 0.9554
 Class 1: 0.9470
 Class 2: 0.7980
 Class 3: 0.7711
 Class 4: 0.8500
 Class 5: 0.8000
 Class 6: 0.9000
 Class 7: 0.8571
 Class 8: 0.7349
 Class 9: 0.9096

It can be seen that the model is most confused about classes 2, 3, 5, and 8.

3.5.2 Confusion Matrix

	0	1	2	3	4	5	6	7	8	9
0	[[343	1	2	2	1	2	4	0	2	2]
1	[2	250	1	0	3	0	5	2	1	0]
2	[12	0	158	10	6	2	1	3	6	0]
3	[9	0	8	128	0	15	0	0	6	0]
4	[7	1	5	1	170	2	3	1	5	5]
5	[9	0	3	10	2	128	1	0	5	2]
6	[4	2	4	1	5	1	153	0	0	0]
7	[0	2	2	2	6	0	0	126	0	9]
8	[10	1	8	4	5	5	3	1	122	7]
9	[2	3	1	1	5	0	0	0	4	161]]

Based on the confusion matrix, we can make the following observations:

1. 2 is most commonly mistaken for 0.
2. 3 is most commonly mistaken for 0 and 5.
3. 5 is most commonly mistaken for 0 and 3.
4. 8 is most commonly mistaken for 0, 2, and 9

3.5.3 Classification Report

Classification Report:

	precision	recall	f1-score	support
0	0.8618	0.9554	0.9062	359
1	0.9615	0.9470	0.9542	264

2	0.8229	0.7980	0.8103	198
3	0.8050	0.7711	0.7877	166
4	0.8374	0.8500	0.8437	200
5	0.8258	0.8000	0.8127	160
6	0.9000	0.9000	0.9000	170
7	0.9474	0.8571	0.9000	147
8	0.8079	0.7349	0.7697	166
9	0.8656	0.9096	0.8871	177
accuracy			0.8665	2007
macro avg	0.8635	0.8523	0.8571	2007
weighted avg	0.8665	0.8665	0.8656	2007

- Digits such as 0 (95.54% recall), 1 (94.70%), and 6 (90.00%) were recognized with high precision and recall. These digits have distinctive and consistent shapes, which HOG is well-suited to represent.
- Digits like 2, 3, 5, and 8 had lower recall values (especially 8 with 73.49%), indicating that there is confusion with similar-shaped digits. It is difficult for HOG to distinguish global structures in these cases.
- Most digits have reasonably balanced precision and recall, showing that HOG features lead to stable predictions without overfitting to a specific class.
- The macro F1-score (85.71%) and weighted F1-score (86.65%) are close, which suggests that performance is consistent between more frequent and less frequent classes.

3.6 Final Pipeline Results

Our fourth and final pipeline, which included the HOG, zoning, and projection histogram features showed improvement with numbers that the model was previously confused about. It yielded the following results:

3.6.1 Accuracy

Overall Accuracy: 0.8969

Per-Class Accuracy:

Class 0: 0.9610
Class 1: 0.9545
Class 2: 0.9091
Class 3: 0.8253
Class 4: 0.9150
Class 5: 0.7625
Class 6: 0.9294
Class 7: 0.8844
Class 8: 0.7831
Class 9: 0.9209

It can be seen that the addition of the projection histogram improved the accuracy of the classes that the model was confused about before (2 (+11.11%), 3 (+5.42%), and 4 (+6.50%)), and other digits also increased slightly. However, there is a decrease in accuracy for class 5 (-3.75%). This suggests that projection-based features may have introduced ambiguity for that class.

3.6.2 Confusion Matrix

	0	1	2	3	4	5	6	7	8	9
0	[[345	0	4	3	1	1	3	1	1	0]
1	[0	252	4	0	4	0	3	1	0	0]
2	[8	0	180	5	2	2	0	0	1	0]
3	[6	0	4	137	0	11	0	1	7	0]
4	[3	1	4	1	183	3	2	0	1	2]
5	[12	0	7	12	1	122	2	0	1	3]
6	[4	0	3	0	2	2	158	0	1	0]
7	[1	1	1	0	8	0	0	130	1	5]
8	[9	1	6	7	5	2	4	0	130	2]
9	[2	1	2	1	2	0	0	3	3	163]]

Based on the confusion matrix, we can make the following observations:

1. We can see that the diagonal entries increased across most classes, so more digits were classified correctly.
2. Many off-diagonal errors decreased noticeably, mostly for digits 2 (-22 errors), 3(-9), and 4 (-13).
3. Digit 5 (+6 errors) had a slight drop in accuracy. Which may be due to tighter boundaries causing more overlap with similar shapes.

3.6.3 Classification Report

Classification Report:

	precision	recall	f1-score	support
0	0.8846	0.9610	0.9212	359
1	0.9844	0.9545	0.9692	264
2	0.8372	0.9091	0.8717	198
3	0.8253	0.8253	0.8253	166
4	0.8798	0.9150	0.8971	200
5	0.8531	0.7625	0.8053	160
6	0.9186	0.9294	0.9240	170
7	0.9559	0.8844	0.9187	147
8	0.8904	0.7831	0.8333	166
9	0.9314	0.9209	0.9261	177
accuracy			0.8969	2007
macro avg	0.8961	0.8845	0.8892	2007
weighted avg	0.8979	0.8969	0.8963	2007

- Our final pipeline had a 3.04% increase in accuracy compared to our first pipeline which only used HOG features.
- Macro averages improve consistently. This shows that performance increases across all digits.
- Digits 2 and 4 showed the largest improvements (+11.11% and +6.50% recall respectively).
- Digit 5 had a small drop in recall (80.0% → 76.25%) but precision improved. This means that although the model is classifying fewer 5s, it is with more certainty.

4 Discussion and Conclusion

In this paper, we explored the effectiveness of image processing and statistical descriptors for handwritten digit recognition.

Our initial baseline used only Histogram of Oriented Gradients (HOG), achieving an overall accuracy of 86.35%. While this demonstrated strong performance on clearly distinguishable digits such as 0,

1, and 6, it struggled with structurally similar digits such as 2, 3, 5, and 8, as shown by the confusion matrix and per-class accuracy metrics.

To address these weaknesses, we incorporated additional spatial descriptors. Zoning added regional intensity information, and projection histograms provided global structural context by summarizing ink distribution along rows and columns. These additions led to a final pipeline that significantly improved the classification of previously confused digits, particularly:

- 2: recall increased by +11.11%
- 3: recall increased by +5.42%
- 4: recall increased by +6.50%

The overall test accuracy rose to 89.69%, and macro-averaged precision, recall, and F1-score all improved as well. The increased accuracy confirms the value of combining local gradient features (HOG) with global spatial features (projection histograms) in classical pipelines.

Slight performance degradation occurred for digit 5, whose recall decreased by 3.75%. The recall for digit 5 likely decreased due to feature overlap with digits like 3 and 8, or possibly noise introduced by global features (projection histograms) that don't distinguish 5-specific curvature well.

It can be seen that adding zoning and projection histograms helped separate digit classes with overlapping gradient profiles. The final pipeline provides a balanced, interpretable, and stronger feature representation.

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